

Sri Lanka Institute of Information Technology



Lab sheet 3

2026-Y2-S1-KU-19

Member Name	Registration No.	Assigned Function
Jayawardena S.D.D.L.	IT25101719	User Registration & Account Management
Madugalle E.W.M.W.W.S.A.K.	IT25101711	Product Listing Management
Alahakoon A.R.S.S.	IT25101690	Search & Browse / Product Catalog
Sooriyaarachchi A.G.S.T.	IT25101736	Shopping Cart & Checkout
Jayasinghe J.M.N.R.N.	IT25101657	Order Management & Tracking
Ahamed M. R. S.	IT25101678	Admin Management Panel

Web based Market Store

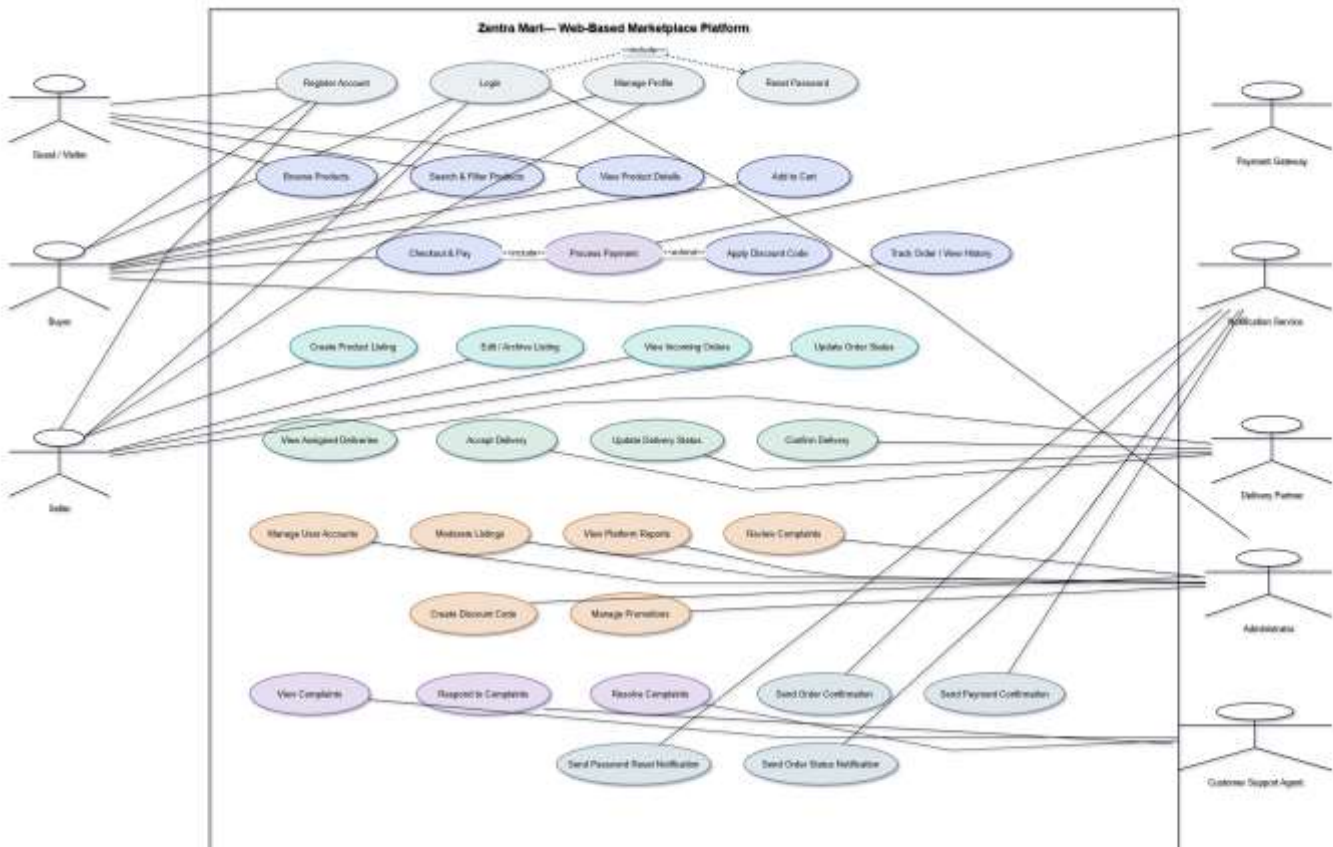
Zentra Mart

Software Engineering | SE2030

B.Sc. (Hons) in Information Technology

Complete Use Case Diagram

The diagram models the complete Zentra Mart system, including all identified human and external actors and their interactions with the system.



- **System:** The diagram represents the complete Zentra Mart Web-Based Marketplace Platform.

- Human Actors: Guest/Visitor, Buyer, Seller, Administrator, Customer Support Agent, and Delivery Partner.
- External Actors: Payment Gateway and Notification Service.
- Account & Authentication: Includes Register Account, Login, Manage Profile, and Reset Password.
- Buyer Functions: Includes browsing products, searching and filtering products, viewing product details, managing the cart, checkout and payment, applying discount codes, and tracking orders.
- Seller Functions: Includes creating and managing product listings, viewing incoming orders, and updating order statuses.
- Administration: Includes managing user accounts, moderating listings, viewing platform reports, reviewing complaints, and managing promotions and discount codes.
- Delivery Management: Includes viewing assigned deliveries, accepting deliveries, updating delivery status, and confirming deliveries.
- Customer Support: Includes viewing, responding to, and resolving customer complaints.
- Payment Processing: Checkout & Pay <<include>> Process Payment, with the Payment Gateway acting as the external payment service.
- Optional Discount: Apply Discount Code <<extend>> Checkout & Pay, because applying a discount is optional.
- Notifications: The Notification Service handles system notifications such as order confirmations, payment confirmations, order-status notifications, and password-reset notifications.

Individual Use Case Scenario

Member 1

Member Name	Jayawardena S.D.D.L.
Registration No.	IT25101719
Use Case ID	UC-01
Use Case Name	Register Account
Primary Actor	Guest / Visitor
Secondary Actor(s)	Notification Service

Description

This use case describes how a new visitor creates an account on Zentra Mart selecting the role of Buyer or Seller, so they can access role-based features of the platform.

Pre-condition

The visitor has access to the Zentra Mart platform and does not already hold an account.

Main Flow

1. Visitor selects “Sign Up” on the platform.
2. Visitor enters their name, email, mobile number, and password.
3. Visitor selects their role as Buyer or Seller.
4. System validates the submitted details.
5. System creates a new user account with the selected role.
6. System requests the Notification Service to send a confirmation OTP to the visitor’s email or mobile.
7. Visitor enters the OTP to confirm the account.
8. System activates the account and displays the appropriate role-based dashboard.

Extensions

2.a. If required fields are missing or invalid:

- (a) Display a validation error message.
- (b) Allow the visitor to correct and resubmit the form.

4.a. If the email is already registered:

- (a) Display an “account already exists” message.
- (b) Redirect the visitor to the Login use case.

7.a. If the OTP is incorrect or expired:

- (a) Display an error message.

- (b) Allow the visitor to request a new OTP.

Member 2

Member Name	Madugalle E.W.M.W.W.S.A.K.
Registration No.	IT25101711
Use Case ID	UC-02
Use Case Name	Create Product Listing
Primary Actor	Seller
Secondary Actor(s)	None

Description

This use case describes how a seller adds a new product to the marketplace catalog, including product details, images, price, and stock, so buyers can discover and purchase it.

Pre-condition

The seller has an active, verified account and is logged into the platform.

Post-condition

A new active product listing exists and is visible in the buyer-facing catalog

Main Flow

1. Logged-in seller selects “Add New Listing”.
2. Seller enters the product name, description, price, stock quantity, and category.
3. Seller uploads one or more product images.
4. System validates the listing details and images.
5. System saves the listing with status “Active”.
6. System publishes the active listing to the buyer-facing catalog.

Extensions

2.a. If required fields are missing or invalid:

- (a) Display a validation error message.
- (b) Allow the seller to correct and resubmit the form.

3.a. If an uploaded image exceeds the size or format limit:

- (a) Display an error message.

(b) Allow the seller to re-upload a valid image.

Member 3

Member Name	Alahakoon A.R.S.S.
Registration No.	IT25101690
Use Case ID	UC-03
Use Case Name	Search & Filter Products
Primary Actor	Buyer
Secondary Actor(s)	Guest / Visitor

Description

This use case describes how a buyer or guest searches, filters, and sorts the product catalog to find relevant active products.

Pre-condition

The buyer or guest visitor has access to the Zentra Mart platform and active product listings exist in the catalog.

Post-condition

The system displays the filtered and sorted product results; the user may proceed to the separate View Product Details use case.

Main Flow

1. Buyer or guest visitor enters a keyword in the search bar.
2. User applies optional filters such as category and price range.
3. System queries the product catalog for matching active listings.
4. System displays matching products with price and thumbnail.
5. User selects a sort order such as price, newest, or rating.
6. System re-orders the results accordingly.
7. Use case ends when the filtered and sorted results are displayed.

Extensions

3.a. If no products match the search or filter criteria:

- (a) Display a “no results found” message.
- (b) Suggest that the user broaden the search.

5.a. If the user applies an invalid filter combination:

- (a) Display a warning message.
- (b) Reset the filters to their default values.

Member 4

Member Name	Sooriyaarachchi A.G.S.T.
Registration No.	IT25101736
Use Case ID	UC-04
Use Case Name	Checkout & Pay
Primary Actor	Buyer
Secondary Actor(s)	Payment Gateway, Notification Service

Description

This use case describes how a buyer completes a purchase from the shopping cart, including delivery confirmation, payment processing, optional discount application, order creation, and order confirmation.

Pre-condition

The buyer is logged in and has at least one item in the shopping cart.

Post-condition

A confirmed order exists, stock levels are updated, and the buyer has received an order confirmation.

Main Flow

1. Buyer selects “Checkout & Pay” from the shopping cart.
2. System displays the order summary with item prices and total.

3. Buyer confirms the delivery details.
4. Buyer optionally applies a valid discount code.
5. Buyer selects a payment method and submits payment.
6. System reserves stock and invokes the Process Payment use case.
7. Payment Gateway returns the payment result.
8. System creates the order, updates stock, and requests the Notification Service to send an order confirmation.
9. Use case ends when the order confirmation is displayed to the buyer.

Extensions

1.a. If the cart is empty:

- (a) Display a message prompting the buyer to add items first.

4.a. If the discount code is invalid or expired:

- (a) Display an error message.
- (b) Continue checkout without applying the discount.

5.a. If required payment details are missing or invalid:

- (a) Display a validation error message.
- (b) Allow the buyer to correct and resubmit.

7.a. If payment verification fails:

- (a) Display a payment failed message.
- (b) Release the reserved stock.
- (c) Allow the buyer to retry payment.

Member 5

Member Name	Jayasinghe J.M.N.R.N.
Registration No.	IT25101657
Use Case ID	UC-05
Use Case Name	Update Order Status
Primary Actor	Seller
Secondary Actor(s)	Notification Service

Description

This use case describes how a seller processes and updates the status of an incoming order as it moves through fulfilment, while the buyer is informed through the notification service.

Pre-condition

A confirmed order exists and is awaiting fulfilment by the seller.

Post-condition

The order status is updated with a timestamp and the Notification Service has been requested to notify the buyer.

Main Flow

1. Seller opens “Incoming Orders” on the seller dashboard.
2. System displays orders awaiting fulfilment.
3. Seller selects an order to process.
4. Seller updates the order status, for example to “Shipped”.
5. System validates and records the status change with a timestamp.
6. System requests the Notification Service to send the buyer an order-status notification.
7. Use case ends when the updated status is stored and the notification request is completed.

Extensions

3.a. If the selected order has already been cancelled:

(a) Display a message indicating the order is no longer active.

4.a. If the seller attempts an invalid status transition, such as “Delivered” before “Shipped”:

- (a) Display an error message.
- (b) Keep the previous status unchanged.

Member 6

Member Name	Ahamed M. R. S.
Registration No.	IT25101678
Use Case ID	UC-06
Use Case Name	Moderate Listings
Primary Actor	Administrator
Secondary Actor(s)	Notification Service

Description

This use case describes how an administrator reviews product listings that have been flagged or reported, deciding whether to approve or remove them to keep the marketplace safe and trustworthy.

Pre-condition

At least one product listing has been flagged for review or reported by a user.

Post-condition

The listing status is updated to approved or removed, and the Notification Service has been requested to notify the relevant seller of the decision.

Main Flow

1. Administrator opens the “Listings” section of the admin panel.
2. System displays listings flagged for review or reported by users.
3. Administrator reviews a flagged listing’s details.
4. Administrator decides to approve or remove the listing.
5. System updates the listing status accordingly.
6. If required, system records the decision and requests the Notification Service to notify the relevant seller.
7. Use case ends when the updated listing status is reflected across the platform.

Extensions

3.a. If the listing was already removed by the seller:

- (a) Display a message indicating that the listing no longer exists.
- (b) Remove it from the review queue.

4.a. If the administrator removes the listing:

- (a) Record the reason for removal in the audit trail.
- (b) Include the removal reason in the notification request to the Notification Service.